



SYDNEY BOYS HIGH SCHOOL
MOORE PARK, SURRY HILLS

2012

**HSC ASSESSMENT
TASK #1**

**Mathematics
Extension 2**

General Instructions

- Reading time – 5 minutes.
- Working time – 90 minutes.
- Write using black or blue pen.
- Board approved calculators may be used.
- All necessary working should be shown in every question if full marks are to be awarded.
- Marks may **NOT** be awarded for messy or badly arranged work.
- Answers should be in simplest exact form unless specified otherwise.
- Start each **NEW** section in a separate answer booklet.
- Each section is to be returned in a separate bundle.

Total Marks - 88

- Attempt Questions 1 - 6
- All questions are NOT of equal value.

Examiner: *A. Fuller*

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax,$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x, \quad x > 0$

Section A

Question 1 (15 marks)

- (a) Find $\int \frac{5}{\cos^2 x} dx$. **1**
- (b) Find the exact value of the following: **4**
- (i) $\cos^{-1}\left(\frac{\sqrt{3}}{2}\right)$
- (ii) $\tan^{-1}\left(\tan \frac{5\pi}{6}\right)$
- (iii) $\sin\left(2 \sin^{-1}\left(\frac{1}{\sqrt{3}}\right)\right)$.
- (c) Write the following in the form $a + ib$, where a and b are real: **6**
- (i) $\overline{3 - 4i}$
- (ii) $\frac{1}{3 - 4i}$
- (iii) the two square roots of $3 - 4i$.
- (d) (i) Express the following in the form $r(\cos \theta + i \sin \theta)$, **4**
where $r > 0$ and $-\pi < \theta \leq \pi$.
- (α) $\sqrt{3} - i$
- (β) $(\sqrt{3} - i)^7$
- (ii) Hence, or otherwise, write $(\sqrt{3} - i)^7$ in the form $x + iy$,
where x and y are real.

Question 2 (15 marks)

(a) $P(x) = 3x^3 - 5x^2 + 4x + 2.$ 4

(i) Show that $(1 + i)$ is a root of $P(x) = 0$.

(ii) Explain why $(1 - i)$ is also a root of $P(x) = 0$.

(iii) Hence, or otherwise, factorise $P(x)$ over the Real field.

(b) Evaluate $\int_1^4 |2 - x| dx.$ 2

(c) (i) Show that $\frac{1}{4+5 \sin^2 x} = \frac{2}{13-5 \cos 2x}.$ 4

(ii) Hence, or otherwise, find $\int \frac{dx}{4+5 \sin^2 x}.$

(d) Sketch the locus of the following on separate argand diagrams: 5

(i) $|z + i| \leq 1$

(ii) $\Re(z + iz) < 1$

(iii) $2|z| = z + \bar{z} + 4$

Section B (Use a SEPARATE writing booklet)

Question 3 (18 marks)

(a) If α, β, γ are the roots of the polynomial equation $2x^3 - 3x + 1 = 0$. **6**

(i) Find the value of the following:

(α) $\alpha\beta\gamma$

(β) $(1 - \alpha)(1 - \beta)(1 - \gamma)$

(γ) $\alpha^2 + \beta^2 + \gamma^2$

(δ) $\alpha^4 + \beta^4 + \gamma^4$

(ii) Form a polynomial equation which has roots $\frac{1}{2\alpha+\beta+\gamma}, \frac{1}{\alpha+2\beta+\gamma}, \frac{1}{\alpha+\beta+2\gamma}$.

(b) (i) Write $\frac{2x^2+x+5}{(x-3)(x^2+4)}$ in the form $\frac{A}{x-3} + \frac{Bx+C}{x^2+4}$. **5**

(ii) Hence, or otherwise, find $\int \frac{2x^2+x+5}{(x-3)(x^2+4)} dx$.

(c) (i) State the domain and range of $y = \sin^{-1}\left(\frac{1}{x}\right), x > 0$. **7**

(ii) Find $\frac{dy}{dx}$.

(iii) Show that $\int \frac{dx}{x\sqrt{x^2-1}} = \sec^{-1} x + C$ using the substitution $x = \sec \theta$.

(iv) Consider $y = \sec^{-1} x$ to be $x = \sec y$ for $0 \leq y < \frac{\pi}{2}$.

Using the results from (ii) and (iii) write $\sec^{-1} x$ in terms of $\sin^{-1}\left(\frac{1}{x}\right)$.

Question 4 (15 marks)

(a) Given that $|z_1| = 15$ and $z_2 = -3 + 4i$. 4

(i) Find the maximum value of $|z_1 + z_2|$.

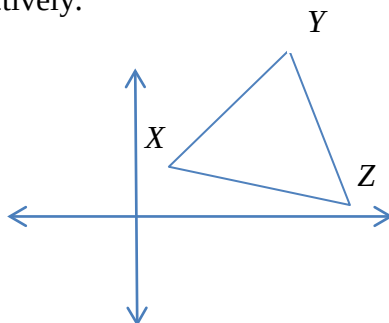
(ii) Hence, find z_1 if $|z_1 + z_2|$ takes its maximum value.

(b) (i) Show that $\int_0^\pi \frac{\sin x}{\sqrt{1+\cos^2 x}} dx = 2 \ln(1 + \sqrt{2})$ 5

using the substitution $u = \cos x$.

(ii) Hence, or otherwise, evaluate $\int_0^\pi \frac{x \sin x}{\sqrt{1+\cos^2 x}} dx$.

(c) In the diagram below, the points X , Y and Z correspond to the complex numbers x , y and z respectively. 6



Find the complex numbers represented by:

(i) the vector OX (where O is the origin)

(ii) the vector XZ

(iii) the point A such that $XYAZ$ is a parallelogram

(iv) the point C , the centroid of $\triangle XYZ$.

Note: The centroid of a triangle is the point of intersection of the three medians.

You may assume that the centroid lies two-thirds along a median from the vertex.

Section C (Use a SEPARATE writing booklet)

Question 5 (13 marks)

- (a) Let α be the complex root of the polynomial equation $z^7 = 1$ with **7**

the smallest positive argument.

Let $\theta = \alpha + \alpha^2 + \alpha^4$ and $\phi = \alpha^3 + \alpha^5 + \alpha^6$.

- (i) Explain why $\alpha^7 = 1$ and $\alpha^6 + \alpha^5 + \alpha^4 + \alpha^3 + \alpha^2 + \alpha + 1 = 0$.

- (ii) Show that $\theta + \phi = -1$ and $\theta\phi = 2$.

- (iii) Show that $\theta = -\frac{1}{2} + i\frac{\sqrt{7}}{2}$ and $\phi = -\frac{1}{2} - i\frac{\sqrt{7}}{2}$.

- (iv) Show that $-\cos\frac{\pi}{7} + \cos\frac{2\pi}{7} + \cos\frac{4\pi}{7} = -\frac{1}{2}$.

- (b) 4 students have yet to be placed in a sport. **6**

There are 6 different sports to choose from. How many ways can this be done if:

- (i) there are no restrictions
- (ii) they must each be placed in different sports
- (iii) no more than 2 can be placed in the same sport
- (iv) 2 particular students can't play the same sport?

Question 6 (12 marks)

(a) $I_n = \int_0^1 x^n \sqrt{1-x} \, dx$

8

(i) Show that $I_n = \frac{2n}{2n+3} I_{n-1}$.

(ii) Use mathematical induction to prove that $I_n = \frac{n!(n+1)!}{(2n+3)!} 4^{n+1}$

for positive integers n .

(iii) Hence, find I_3 .

(b) Prove that $ax^3 + 3bx^2 + 3cx + d$ has a triple zero if a, b, c, d are

4

successive terms of a geometric series.

End of paper

EXT 2 - Section A 2012
Half-Yearly

1. (a) $\int \frac{5}{\cos^2 x} dx$

$= 5 \int \sec^2 x dx$

$= 5 \tan x + C$ ✓ (1)

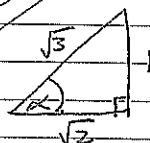
(b) (i) $\cos^{-1}\left(\frac{\sqrt{3}}{2}\right) = \frac{\pi}{6}$ ✓ (1)

(ii) $\tan^{-1}\left(\tan \frac{5\pi}{6}\right) = \frac{\pi}{6}$ ✓ (1)

(iii) $\sin\left(2 \sin^{-1}\left(\frac{1}{\sqrt{3}}\right)\right)$

$= \sin(2\alpha)$ where $\alpha = \sin^{-1}\left(\frac{1}{\sqrt{3}}\right)$

$\sin \alpha = \frac{1}{\sqrt{3}}$



$= 2 \sin \alpha \cos \alpha$

$= 2 \cdot \frac{1}{\sqrt{3}} \cdot \frac{\sqrt{2}}{\sqrt{3}}$

$= \frac{2\sqrt{2}}{3}$ ✓ (2)

(c) (i) $3-4i = B+4i$ ✓ (1)

(ii) $\frac{1}{3-4i} = \frac{1}{3-4i} \times \frac{3+4i}{3+4i} = \frac{3+4i}{25}$ ✓ (1)

(iii) $\sqrt{3-4i} = a+bi$ ✓

$3-4i = (a+bi)^2$

$3-4i = a^2-b^2+2abi$

$\Rightarrow 3 = a^2-b^2$ (1) ✓

$-4 = 2ab$ (2)

From (2) $b = \frac{-2}{a}$

Sub m (1)

$\Rightarrow 3 = a^2 - \frac{4}{a^2}$

$3a^2 = a^4 - 4$

$a^4 - 3a^2 - 4 = 0$

$(a^2-4)(a^2+1) = 0$

1 (c) (iii) cont $a = \pm 2$ or $a = \pm \sqrt{1} \rightarrow$ reject since $a, b \in \mathbb{R}$

When $a=2, b=-1$ ✓

$a=-2, b=1$

$\therefore \sqrt[4]{3-4i} = 2-i$ or $-2+i$ (4)

1 (d) (i) (a) $\sqrt{3}-i$

Then $r = \sqrt{3+1} = 2$

and $\theta = \tan^{-1}\left(\frac{-1}{\sqrt{3}}\right) = -\frac{\pi}{6}$

$\therefore \sqrt{3}-i = 2\left(\cos\left(-\frac{\pi}{6}\right) + i \sin\left(-\frac{\pi}{6}\right)\right)$ ✓ (1)

(b) $(\sqrt{3}-i)^7 = (r \cos \theta)^7$

$= \left[2 \cos\left(-\frac{\pi}{6}\right)\right]^7$

$= 2^7 \cos\left(7 \times -\frac{\pi}{6}\right)$ by De Moivre's Th.

$= 128 \cos\left(-\frac{7\pi}{6}\right)$ ✓

$= 128 \cos \frac{5\pi}{6}$

$= 128 \left(\cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6}\right)$ ✓ (2)

$$\begin{aligned}
 1. (d) (ii) \quad (\sqrt{3}-i)^7 &= 128 \left(\cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6} \right) \\
 &= 128 \left(-\frac{\sqrt{3}}{2} + i \cdot \frac{1}{2} \right) \\
 &= 64(-\sqrt{3} + i) \\
 &= -64\sqrt{3} + 64i \quad \checkmark \quad (1)
 \end{aligned}$$

Q2 $P(x) = 3x^3 - 5x^2 + 4x + 2$

(i) Let $x = 1+i$.

$$x^2 = (1+i)^2 = 1 - 1 + 2i = 2i$$

$$x^3 = 2i(1+i) = 2i - 2$$

$$\begin{aligned}
 P(1+i) &= 3(2i-2) - 5(2i) + 4(1+i) + 2 \quad (1) \\
 &= 6i - 6 - 10i + 4 + 4i + 2 \\
 &= 0
 \end{aligned}$$

(ii) By the conjugate root theorem if $(a+ib)$ is a complex root of $P(x)$ where $P(x)$ has real coefficients then $(a-ib)$ is also a root. $\checkmark \quad (1)$

(iii) Then $(x-(1-i))(x-(1+i))$ is a factor of $P(x)$

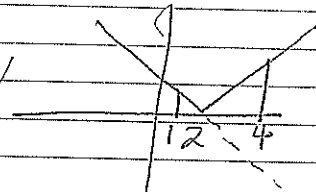
$$\Rightarrow x^2 - x(1+i) - x(1-i) + 2 \text{ is a factor}$$

$$\Rightarrow x^2 - 2x + 2 \text{ is a factor.} \quad \checkmark$$

$$3x^3 - 5x^2 + 4x + 2 = (x^2 - 2x + 2)(3x+1) \quad (2)$$

by observation

$$\therefore P(x) = (x^2 - 2x + 2)(3x+1) \text{ over } \mathbb{R} \quad \checkmark$$

$$\begin{aligned}
 2(b) \quad \int_1^4 |2-x| dx &= \int_1^2 (2-x) dx + \int_2^4 (x-2) dx \quad \checkmark \\
 &= \left[2x - \frac{x^2}{2} \right]_1^2 + \left[\frac{x^2}{2} - 2x \right]_2^4 \\
 &= \left[(4-2) - (2-\frac{1}{2}) \right] + \left[(8-8) - (2-4) \right] \\
 &= \frac{1}{2} \quad \checkmark \quad + 2 \quad (2) \\
 &= 2\frac{1}{2} \text{ or } 5/2
 \end{aligned}$$


(c) (i) Show $\frac{1}{4+5\sin^2 x} = \frac{2}{13-5\cos 2x}$

$$RHS = \frac{2}{13-5\cos 2x}$$

$$= \frac{2}{13-5(2\cos^2 x - 1)} \quad \checkmark$$

$$= \frac{2}{13-10\cos^2 x + 5}$$

$$= \frac{2}{18-10\cos^2 x}$$

$$= \frac{1}{9-5(1-\sin^2 x)} \quad (2)$$

$$= \frac{1}{9-5+5\sin^2 x} \quad \checkmark$$

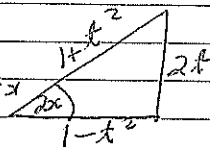
$$= \frac{1}{4+5\sin^2 x} = LHS \quad \#$$

$$2(c)(ii) \int \frac{dx}{4+5\sin^2 x}$$

$$= 2 \int \frac{1}{13-5\cos 2x} dx$$

$$= 2 \int \frac{1}{-5\cos 2x + 13} dx$$

Let $t = \tan x$
 $\frac{dt}{dx} = \sec^2 x = 1 + \tan^2 x$
 $dx = \frac{dt}{1+t^2}$



$$= 2 \int \frac{1}{-5 \times \frac{1-t^2}{1+t^2} + 13} \cdot \frac{dt}{1+t^2}$$

$$= 2 \int \frac{dt}{-5(1-t^2) + 13(1+t^2)}$$

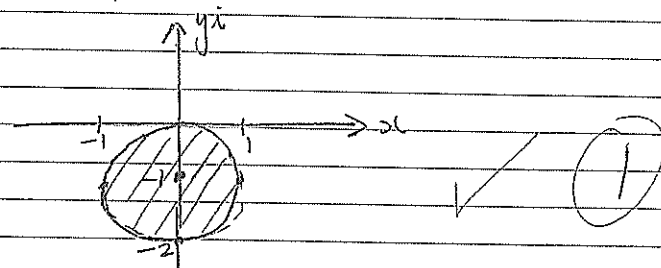
$$= 2 \int \frac{dt}{-5 + 5t^2 + 13 + 13t^2} = 2 \int \frac{dt}{18t^2 + 8}$$

$$= \int \frac{dt}{4 + 9t^2} = \int \frac{dt}{4 + (3t)^2}$$

$$= \frac{1}{3} \cdot \frac{1}{2} \tan^{-1} \left(\frac{3t}{2} \right) + C$$

$$= \frac{1}{6} \tan^{-1} \left(\frac{3}{2} \tan x \right) + C$$

$$2(d)(i) |z+i| \leq 1$$



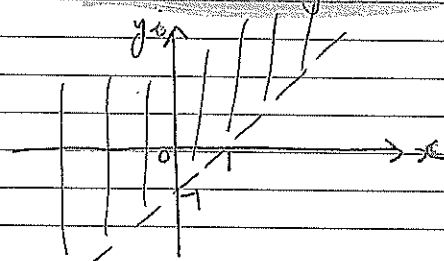
$$(ii) \operatorname{Re}(z+iz) < 1 \quad \text{Let } z = x+yi$$

$$\Rightarrow \operatorname{Re}(x+yi + i(x+yi))$$

$$= \operatorname{Re}(x-y + (x+y)i)$$

$$= x-y$$

$$\therefore \text{sketch } x-y < 1$$



$$(iii) 2|z| = z + \bar{z} + 4$$

$$\text{let } z = x+yi$$

$$2\sqrt{x^2+y^2} = x+yi + x-yi + 4$$

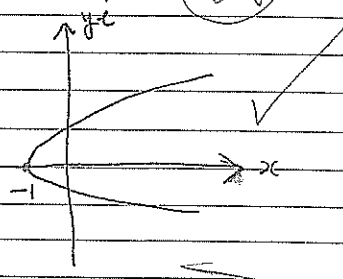
$$2\sqrt{x^2+y^2} = 2x+4$$

$$\sqrt{x^2+y^2} = x+2$$

$$x^2+y^2 = x^2+4x+4$$

$$\Rightarrow y^2 = 4(x+1)$$

Parabola V(-1,0)



SECTION B

$$(a) (i) (\alpha) \quad \alpha\beta r = -\frac{d}{a} \\ = -\frac{1}{2}$$

$$(p) (1-\alpha)(1-\beta)(1-r)$$

$$= (1-\alpha)(1-r-\beta+\beta r)$$

$$= (1-r-\beta+\beta r - \alpha + \alpha r + \alpha\beta - \alpha\beta r)$$

$$= 1 - (\alpha + \beta + r) + (\alpha\beta + \beta r + \alpha r) - \alpha\beta r$$

$$= 1 - 0 = \frac{3}{2} + \frac{1}{2}$$

$$= \textcircled{0}$$

$$(r) (\alpha^2 + \beta + r)^2 = (\alpha^2 + \beta^2 + r^2) + 2(\alpha\beta + \beta r + \alpha r)$$

$$\alpha^2 + \beta^2 + r^2 = (\alpha + \beta + r)^2 - 2(\alpha\beta + \beta r + \alpha r)$$

$$= -2\left(-\frac{3}{2}\right)$$

$$= +3$$

$$(s) 2x^4 - 3x^2 + x = 0$$

$$2(\alpha^4 + \beta^4 + r^4) - 3(\alpha^2 + \beta^2 + r^2) + (\alpha + \beta + r) = 0$$

$$=$$

$$2(\alpha^4 + \beta^4 + r^4) = 3(\alpha^2 + \beta^2 + r^2) + (\alpha + \beta + r)$$

$$= 3(+3) - 0$$

$$\alpha^4 + \beta^4 + r^4 = +9 \\ = +\frac{9}{2}$$

$$ii) \frac{1}{2\alpha + \beta + r} ; \frac{1}{\alpha + 2\beta + r} ; \frac{1}{\alpha + \beta + 2r}.$$

$$= \frac{1}{\alpha + (\alpha + \beta + r)} ; \frac{1}{\beta + (\alpha + \beta + r)} ; \frac{1}{r + (\alpha + \beta + r)}$$

$$= \frac{1}{\alpha} ; \frac{1}{\beta} ; \frac{1}{r}.$$

$$\text{Let } X = \frac{1}{x}.$$

$$x = \frac{1}{X}$$

$$\frac{2}{X^3} - \frac{3}{X} + 1 = 0$$

$$2 - 3X^2 + X^3 = 0.$$

$$X^3 - 3X^2 + 2 = 0.$$

$$(b) (i) \frac{2x^2 + x + 5}{(x-3)(x^2+4)} \equiv \frac{A}{x-3} + \frac{Bx+C}{x^2+4}.$$

$$2x^2 + x + 5 \equiv A(x^2+4) + (Bx+C)(x-3).$$

when $x=3$.

$$18 + 3 + 5 = 13A.$$

$$A = 2.$$

x^2

$$2 \equiv A + B$$

$$B = 0$$

x

$$1 \equiv -3B + C$$

$$C = 1.$$

$$\frac{2}{x-3} + \frac{1}{x^2+4}$$

$$(ii) \int \frac{2}{x-3} + \frac{1}{x^2+4} dx$$

$$= 2 \ln(x-3) + \frac{1}{2} \tan^{-1}\left(\frac{x}{2}\right) + C.$$

(i) Domain: ~~$x \geq 1$~~ $x \geq 1$

Range: ~~$0 \leq y \leq \frac{\pi}{2}$~~ $0 \leq y \leq \frac{\pi}{2}$

$$(ii) \frac{dy}{dx} = \frac{1}{\sqrt{1-(\frac{1}{x})^2}} \cdot x - \frac{1}{x^2}$$

$$= - \frac{1}{\sqrt{x^4 - x^2}}$$

$$= - \frac{1}{x\sqrt{x^2 - 1}}$$

$$(iii) \int \frac{dx}{x\sqrt{x^2-1}}$$

$$x = \sec \theta$$

$$\frac{dx}{d\theta} = \sec \theta \tan \theta.$$

$$dx = \sec \theta \tan \theta \cdot d\theta$$

$$= \int \frac{\sec \theta \tan \theta \cdot d\theta}{\sec \theta \sqrt{\sec^2 \theta - 1}}$$

$$= \int \frac{\tan \theta}{\sqrt{\tan^2 \theta}} d\theta$$

$$= \theta + C.$$

$$= \sec^{-1} x + C$$

$$x = \sec \theta$$

$$\theta = \sec^{-1} x$$

$$(iv) \quad y = \sin^{-1}\left(\frac{1}{x}\right)$$

$$\frac{dy}{dx} = - \frac{1}{x\sqrt{x^2-1}}$$

$$y = - \int \frac{1}{x\sqrt{x^2-1}}$$

$$y = - \sec^{-1} x + C$$

$$\sin^{-1}\left(\frac{1}{x}\right) = C - \sec^{-1} x.$$

$$\text{when } x=1$$

$$\sin^{-1}(1) = C - \sec^{-1} 1$$

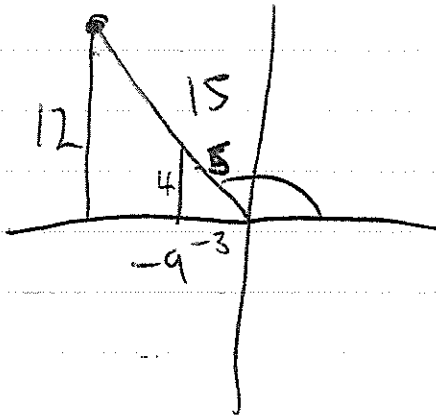
$$\frac{\pi}{2} = C.$$

$$\text{So } \sec^{-1} x = \frac{\pi}{2} - \sin^{-1}\left(\frac{1}{x}\right).$$

QUESTION 4

(a i) $|z_1 + z_2| \leq |z_1| + |z_2|$
 $= 20.$

(ii)



max when $\arg(z_1) = \arg(z_2)$

$$z_1 = -9 + 12i$$

(b i) $\int_0^{\pi} \frac{\sin x}{\sqrt{1+\cos^2 x}} dx$

$$u = \cos x$$

$$-du = \sin x dx$$

$$= \int_{-1}^1 \frac{du}{\sqrt{1+u^2}}$$

when $x=0$ $u=1$
 $x=\pi$ $u=-1$

$$= 2 \int_0^1 \frac{du}{\sqrt{1+u^2}}$$

$$= 2 \left[\ln(u + \sqrt{1+u^2}) \right]_0^1$$

$$= 2 \ln(1 + \sqrt{2})$$

(ii) $I = \int_0^{\pi} \frac{x \sin x}{\sqrt{1+\cos^2 x}} dx$

$$= \int_0^{\pi} \frac{(\pi-x) \sin(\pi-x)}{\sqrt{1+\cos^2(\pi-x)}} dx$$

$$I = +\pi \int_0^\pi \frac{\sin x}{\sqrt{1+\cos^2 x}} dx - \int_0^\pi \frac{x \sin x}{\sqrt{1+\cos^2 x}} dx$$

$$2I = 2\pi \ln(1+\sqrt{2})$$

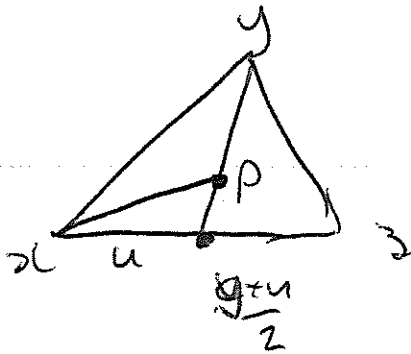
$$I = \pi \ln(1+\sqrt{2})$$

(ci) x

(ii) $z-x$

(iii) $y+z-x$

(iv)



~~$$\begin{aligned}
 u &= \frac{c+y}{2} \\
 p &= \frac{1}{3}(b-u) + u \\
 &= \frac{b}{3} - \frac{u}{3} + \frac{3u}{3} \\
 &= \frac{b}{3} + \frac{c+y}{3}
 \end{aligned}$$~~

$$u = \frac{z+x}{2}$$

$$\begin{aligned}
 p &= \frac{1}{3}(y-u) + u \\
 &= \frac{y}{3} - \frac{u}{3} + \frac{3u}{3} \\
 &= \frac{y}{3} + \frac{z+x}{3} \\
 &= \frac{x+y+z}{3}
 \end{aligned}$$

Q5 (a) (i) $z^7 = 1$

$$\therefore z^7 - 1 = 0$$

$$\therefore (z-1)(z^6 + z^5 + z^4 + z^3 + z^2 + z + 1) = 0$$

now α is a root. $\therefore \boxed{\alpha^7 = 1}$

$$\therefore (\alpha-1)(\alpha^6 + \alpha^5 + \alpha^4 + \alpha^3 + \alpha^2 + \alpha + 1) = 0$$

$$\alpha \neq 1.$$

$$\therefore \boxed{\alpha^6 + \alpha^5 + \alpha^4 + \alpha^3 + \alpha^2 + \alpha + 1 = 0.} \quad (A)$$

(ii)

$$\theta + \phi = \alpha + \alpha^2 + \alpha^4 + \alpha^3 + \alpha^5 + \alpha^6$$

$$= -1 \text{ from (A)}$$

$$\begin{aligned} \theta \phi &= (\alpha + \alpha^2 + \alpha^4)(\alpha^3 + \alpha^5 + \alpha^6) \\ &= \alpha^4 + \alpha^6 + \alpha^7 + \alpha^5 + \alpha^7 + \alpha^8 + \alpha^7 + \alpha^9 + \alpha^{10} \\ &= \alpha^4 + \alpha^6 + 1 + \alpha^5 + 1 + \alpha + 1 + \alpha^2 + \alpha^3 \\ &= 3 + (\alpha + \alpha^2 + \alpha^3 + \alpha^5 + \alpha^5 + \alpha^6) \\ &= 3 + (-1) \\ &= 2. \end{aligned}$$

(iii)

Form an equation (quadratic) with roots θ & ϕ where $\theta + \phi = -1$ & $\theta \phi = 2$

$$x^2 - (\theta + \phi)x + \theta\phi = 0.$$

$$x^2 + x + 2 = 0.$$

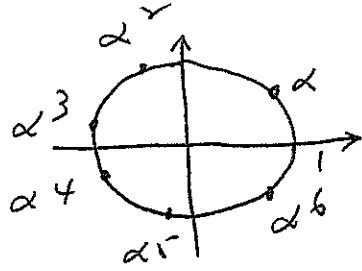
$$x = \frac{-1 \pm \sqrt{1-8}}{2}$$

(CONTD)

$$\therefore x = -\frac{1}{2} \pm i \frac{\sqrt{7}}{2}$$

(2)

Then



Clearly $\theta = \alpha + \alpha^2 + \alpha^4$

has a positive imaginary part.

& $\phi = \alpha^3 + \alpha^5 + \alpha^6$ has a negative imaginary part.

$$\therefore \theta = -\frac{1}{2} + i \frac{\sqrt{7}}{2} \quad \& \quad \phi = -\frac{1}{2} - i \frac{\sqrt{7}}{2}$$

(iv) now $\alpha = \cos \frac{2\pi}{7}$

$$\theta = \alpha + \alpha^2 + \alpha^4 = -\frac{1}{2} + i \frac{\sqrt{7}}{2}$$

$$\text{i.e. } \cos \frac{2\pi}{7} + \cos \frac{4\pi}{7} + \cos \frac{8\pi}{7} = -\frac{1}{2} + i \frac{\sqrt{7}}{2}$$

Taking real parts.

$$\cos \frac{2\pi}{7} + \cos \frac{4\pi}{7} + \cos \frac{8\pi}{7} = -\frac{1}{2}$$

$$\cos \frac{2\pi}{7} + \cos \frac{4\pi}{7} + \cos \pi = -\frac{1}{2}$$

$$\therefore \left| -\cos \frac{\pi}{7} + \cos \frac{2\pi}{7} + \cos \frac{4\pi}{7} = -\frac{1}{2} \right|$$

(3)

$$(b) \quad (i) \quad 6^4 = 1296$$

$$(ii) \quad 6 \times 5 \times 4 \times 3 = 1170.$$

$$(iii) \quad 6^4 - (6 + {}^4C_3 \times 6 \times 5) = 1170.$$

$$\left[\text{OR } 6 \times 5 \times 4 \times 3 + {}^4C_2 \times 6 \times 5 \times 4 + \frac{{}^4C_2 \times 6 \times 5}{2} = 1170 \right]$$

$$(iv) \quad 6^4 - 6^3 = 1080$$

$$\left[\text{OR } 6 \times 5 \times 6 \times 6 = 1080 \right]$$

Prob. (a)

(1)

$$I_n = \int_0^1 x^n \sqrt{1-x} \, dx.$$

$$= \int_0^1 x^n \frac{d}{dx} \left(-\frac{2}{3} \right) (1-x)^{3/2} \, dx.$$

$$= \left[-\frac{2}{3} x^n (1-x)^{3/2} \right]_0^1 + \frac{2}{3} \int_0^1 n x^{n-1} (1-x)^{3/2} \, dx$$

$$= [0-0] + \frac{2}{3} n \int_0^1 x^{n-1} (1-x) (1-x)^{1/2} \, dx.$$

$$= \frac{2n}{3} \int_0^1 x^{n-1} \sqrt{1-x} \, dx = \frac{2n}{3} \int_0^1 x^n \sqrt{1-x} \, dx$$

$$\therefore I_n = \frac{2n}{3} I_{n-1} - \frac{2n}{3} I_n.$$

$$I_n \left(1 + \frac{2n}{3} \right) = \frac{2n}{3} I_{n-1}.$$

$$I_n \frac{2n+3}{3} = \frac{2n}{3} I_{n-1}$$

$$\boxed{I_n = \frac{2n}{2n+3} I_{n-1}}$$

(2)

(b)

Prove
$$I_n = \frac{n! (n+1)!}{(2n+3)!} 4^{n+1}$$

When $n=1$.

$$\begin{aligned} \text{LHS} = I_1 &= \frac{2}{5} I_0 \\ &= \frac{2}{5} \cdot \left[-\frac{2}{3} (1-x)^3 \right]_0^1 \\ &= \frac{2}{5} \times \frac{2}{3} \\ &= \frac{4}{15} \end{aligned}$$

$$\begin{aligned} \text{RHS} &= \frac{1! \times 2!}{5!} \times 4^2 \\ &= \frac{32}{5 \times 4 \times 3 \times 2 \times 1} \\ &= \frac{4}{15} \end{aligned}$$

$\therefore$ True when $n=1$.

Assume the statement to be true when $n=k$.

$$\text{i.e. } I_k = \frac{k! (k+1)!}{(2k+3)!} 4^{k+1}$$

R.T.P. statement is true when $n=k+1$
(using the assumption.)

$$\text{i.e. } I_{k+1} = \frac{(k+1)! (k+2)!}{(2k+5)!} 4^{k+2}$$

(2)

$$\begin{aligned}
\text{now } I_{k+1} &= \frac{2(k+1)}{(2k+5)} \cdot I_k \\
&= \frac{2(k+1)}{2k+5} \times \frac{k! (k+1)!}{(2k+3)!} \times 4^{k+1} \\
&= \frac{2(k+1)}{2k+5} \times \frac{k! \times (k+1)!}{(2k+3)!} \times \frac{2k+4}{2k+4} \times 4^{k+1} \\
&= \frac{4(k+1) k! (k+1)! (k+2)}{(2k+5)!} \times 4^{k+1} \\
&= \frac{(k+1)! (k+2)!}{(2k+5)!} \times 4^{k+2} \\
&= \underline{Q.H.S.}
\end{aligned}$$

$\therefore$ By the Principle of mathematical induction the statement is true for all positive integers.

$$\begin{aligned}
(117) \quad I_3 &= \frac{3! (3+1)! 4^4}{9!} \\
&= \frac{3! \times 4! \times 4^4}{9!} \\
&= \frac{32}{315}
\end{aligned}$$

(5) given $P(x) = ax^3 + 3bx^2 + 3cx + d$.

$$\text{let } a = a$$

$$b = ar$$

$$c = ar^2$$

$$d = ar^3$$

$$\begin{aligned} \therefore P(x) &= ax^3 + 3arx^2 + 3ar^2x + ar^3 \\ &= a(x^3 + 3rx^2 + 3r^2x + r^3) \\ &= a(x+r)^3 \text{ which has} \end{aligned}$$

a triple root of $-r$.

(There are many other ways of doing this question.)